GH3036 Heart Rate Variability Performance Report

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Preface

Purpose

This document is intended to reflect the Heart rate variability performance test result of GH3036.

Audience

This document is intended for:

- Systems and product engineers
- Field Application Engineers
- Test Engineers
- R&D Engineers

Release Notes

This document is the frist official release; the corresponding software version is Package_V00.02.00.00

Revision History

Version	Date	Description
1.0	2025-11-14	The first release, the heart rate variability algorithm is HRV_pre_pv_v1.0.1.0

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1 Config Introduction

Software version: Package_V00.02.00.00, mainly includes:

1. APP version: GHealth_1.0.9.13.apk
2. Algorithm version:

   GH_HR_exc_pv_v1.0.1.0
3. Firmware version: GR551x_VX.X_DEMO_DRV_LIB_GH3036.bin
4. Driver library version: GH(M)3X_DEMO_v0.2.1.0_ALGO.bin

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2 Test Requirements

Table 2-1 Test Requirement Details

Test Scope	Description
Test environment	Office room, gym
Test environment temperature	25°C- 32°C
Testee gender	Male and female
Testee age	20- 40 years old
Testee BMI	18- 30 kg/m^2
Testee skin color level	Levels 2- 5
Reference device	GOOD FRIEND Single-lead dynamic electrocardiogram recorder (DL-191)

Note:

Skin Tone Guide:

The Pantone SkinTone Guide										
Goodix Index	1	2	3	4	5	6	7	8	9	10
Pantone Index	1Y01SP	5Y01SP	2Y06SP	4Y07SP	2Y10SP	3Y11SP	2Y13SP	3R13SP	1R14SP	4R15SP

3 Result

1. Sample quantity: Based on the data quantity recorded in the table.
2. Test method: The testee wears GH3036 device and the reference device with appropriate tightness, then start testing.
3. Test scenario (At Rest): Putting the hands on the desktop without moving during the whole process and collecting the "resting" process data.

3.1 Test results without confidence region

Table 3-1 No confidence region test results

Index / Scene	Sample #	Output time(s)	MAE (ms)	Coefficient of consistency	MAD(Main Device)	MAD(Refrence Device)
Testing	16	7.38	13.65	0.79	30.16	28.75

3.2 Test results with high confidence region

Table 3-2 High confidence region test results

Index / Scene	Sample #	High Confidence %	MAE (ms)	Coefficient of consistency	MAD(Main Device)	MAD(Refrence Device)
Resting	16	98.94%	13.00	0.81	29.94	28.91

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4 Legend

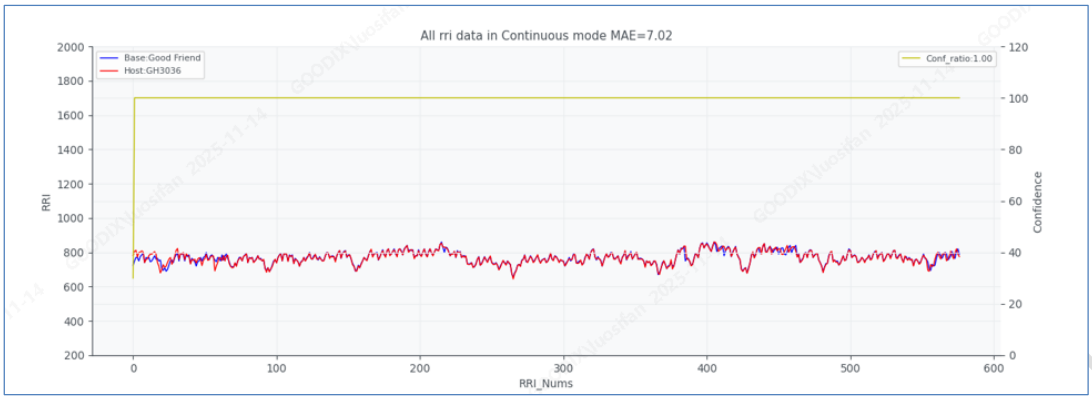

Figure 4-1 Resting

5 Terminology

Table 5-1 Terminology

Indicator	Definition
Output Time	Average value time within 1min.
MAE	Absolute Error, the average absolute error, is defined as, i=1, 2, 3, ... n. In a finite number of measurements, the mean absolute error is often expressed as follows: (∑|di|)/n, where: n is the number of measurements; di is the deviation between a set of measured values and the true value)
Coefficient of consistency	Describes the closeness of the connection between two interval variables, reflecting the strength of the linear correlation between the two variables
MAD	Median absolute deviation, for a single variable data set X1, X2, . . ., Xn, MAD is defined as the median of the absolute deviations from the data point to the median MAD = median (|Xi-median (X)|)
High Confidence %	Data with high confidence level (>80%), or data that algorithm considering highly reliability.

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